Yi Lu

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Education Background

Johns Hopkins University | Whiting School of Engineering | Baltimore, Maryland, USA
Master of Science Degree | 05/2025 | GPA: 3.90/4.00(**top 5%**)

- Major: Data Science Engineering
- Advisor: Prof. Soledad Villar

Chinese University of Hong Kong, Shen Zhen | School of Data Science | Shen Zhen, Guang Dong, China
Bachelor of Science Degree | 07/2024 | GPA: 3.92/4.00(**top 5%**)

- Major: Data Science and Big Data Technology
- Minor & Stream: Statistics, Methodology and Theory
- Advisor: Prof. Andre Mizarek
- Honors: Bowen Scholarship, Academic Performance Scholarship, Dean's List, Mathematical Competition Second Prize

Professional Skills

- Programming Languages: Python, MATLAB, R, Java, SQL, Html, Java Script
- Deep learning platforms: TensorFlow, Pytorch, Yolov
- Languages: Chinese (native speaker), English (professional)

Industry Experience

Angleyes Technology Inc. — Shen Zhen, Guang Dong, China
Algorithm Engineer Intern — 01/2024 - 05/2024

- Honed deep learning platform pytorch basic application, constructing fundamental neural network for prediction
- Refined baseline algorithms from Yolov, configuring code execution environment and attaining executable models; Achieved 70% image recognition accuracy
- Preprocessed images and utilized Roboflow platform for data annotation, obtaining datasets applicable for Yolov training
- Experimented with traffic detection dataset for image recognition, adjusting model parameters and comparing training effects; Accomplished a model with 90% training accuracy
- Collaborated on industrial panels data detection, adjusting batch size and epochs for better training effect; Achieved a model with 95% detection accuracy

Hundun Energy Technology Inc. — Wu Xi, Jiang Su, China
Algorithm Intern — 07/2022 - 08/2022

- Designed basic webpages with certain portals by Flask framework in python
- Conducted data cleaning by pandas package, eliminating outliers and missing values accounting for 10% in original data
- Selected among fundamental machine learning algorithm by real tasks, importing scikit-learn package and implementing basic algorithms to fulfill algorithm functions
- Evaluated baseline models on air-conditioner data, adjusting model parameters and decreasing loss function value by 15%

Research Experience

Graduate Researcher | JHU | Baltimore, Maryland, US | 01/2025 - 08/2025

- Analyzed paper of Scalars Method and Double Pendulum Experiments, exploring the algorithm
- Implemented bilipschitz embedding in pytorch and integrated it into the original
- Compared the prediction results with and without bilipschitz
- Tuning parameters for the multi-layer perceptron network, summarizing a table for prediction results
- Designed further noise addition for the original dataset, testing the bilipschitz embedder effect for model robustness

Research Collaborator | JHU | Baltimore, Maryland, US | 06/2025 - 08/2025

- Preprocessed video data and derived audio, caption, video frames information
- Collaborated with group members for initial answer generation for specific questions
- Evaluated the initial generations by BERT score, STS (Semantic Textual Similarity) and METEOR (Metric for Evaluation of Translation with Explicit Ordering)

Research Assistant | JHU-LCSR | Baltimore, Maryland, US | 08/2025 - present

- Explored **Language Models Trajectory Generator** paper, examining audio-text-motion plan conversion for robot arm
- Implemented GPT-OSS model into the original frame, replacing the original GPT-4o model as an alternative

Undergraduate Researcher | CUHKSZ | Shen Zhen, Guang Dong | 01/2024 - 05/2024
- Investigated paper related to LoRA algorithms, decreasing 90% trainable parameters while preserving training accuracy of LLM
- Referred to GPT-2 pretrained model from Open AI, adopting LoRA algorithm for fine tuning and reinforcing the model's dialogue ability under certain context
- Configured and deployed Conda, CUDA, CUDNN, solving technical issues and ending up with GPU version of Pytorch
- Executed primary model codes and recorded function losses, achieving loss values visualization; Adjusted model parameters and derived model with 15% faster convergence rate
- Implemented LoRA algorithms for fine tuning based on dataset on Hugging Face, increased average dialogue score by 10% and decreased convergent loss function values by 30%

Project Experience

Project Leader | CUHKSZ | Shen Zhen, Guang Dong | 10/2023 - 12/2023
- Assigned tasks and organized weekly meetings, scheduling project timeline and supervising teammates' work; Ensured smooth progress of the whole project
- Implemented Q learning algorithms to train an agent and maximized reward, exploring algorithm to obtain the route with highest cumulative rewards with help of human feedback
- Simulated user interaction and conducted route visualization, tuning human feedback and increasing the reward by 10 times
- Collaborated to accomplish 8-page report to summarize findings and delivered group presentation to demonstrate results

Project Leader | CUHKSZ | Shen Zhen, Guang Dong | 07/2023 - 08/2023
- Generated gaussian, salt-pepper and Moire pattern noise and simulated theoretically noisy space images in Python
- Implemented conventional filters such as mean, median, gaussian filters and innovative filters such as Laplacian, Wiener, Notch filters in Python by cv2 package, applying into noisy images
- Evaluated denoising performance with measure of PSNR (Peak signal-to-noise ratio) and SSIM (Structural Similarity Index Measure), picking filter with best denoising effect; Improved denoising performance by 20% than best conventional filters
- Compared denoising effects of different filters, completing 8-page project report; Attained the best project in class

Publications

Preprint

1. Breaking Forgetting, Training-Free Few-Shot Class-Incremental Learning via Conditional Diffusion
Haidong K., Ketong Q., **Yi L.**
CVPR 2026 submission

2.Generating and Reranking Diverse Answers in Video Question Answering: TREC2025 Video Question Answering Shared Task Description
Dengjia Z.*, Charles W.*, Katherine G., **Yi L.**, Kenton M., Alexander M., Reno K., Benjamin V. D.,
CVPR 2026 submission

Awards and Honors

Bowen Scholarship	2021-2023
Academic Performance Scholarship	2021-2022
Dean’s List	2021-2024
University Students’ Mathematics Competition Second Class Prize	2021-2022